Few-Shot Class-Incremental SAR Target Recognition Based on Hierarchical Embedding and Incremental Evolutionary Network

Li Wang, Xinyao Yang, Haoyue Tan, Xueru Bai, *Senior Member, IEEE*, and Feng Zhou, *Member, IEEE*

***Abstract*— It is difficult to realize effective synthetic aperture radar (SAR) automatic target recognition (ATR) in open scenarios because the ATR model cannot continuously learn from new classes with limited training samples. When adding new classes to the previously trained model, the capability of recognizing old classes may lose due to severe overfitting. To tackle this problem, a few-shot class-incremental SAR ATR method, namely, hierarchical embedding and incremental evolutionary network (HEIEN), is proposed in this article. First, a hierarchical embedding network and a hybrid distance-based classifier are constructed for basic feature extraction and classification. Then, in order to obtain more accurate decision boundaries, an adaptive class-incremental learning (ACIL) module is designed to adjust the weights of classifiers in all tasks by collecting context information from the past to the present. Finally, a pseudo-incremental training strategy is designed to enable effective model training with only a few samples. Experimental results on the moving and stationary target acquisition and recognition (MSTAR) benchmark data set have illustrated that HEIEN performs well with remarkable advantages in few-shot class-incremental SAR ATR tasks.**

***Index Terms*— Automatic target recognition (ATR), class-incremental learning (CIL), few-shot learning (FSL), synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave imaging sensor that can obtain the structure, size, and scattering information of targets in all-day and all-weather scenarios. Therefore, it has found wide applications in various tasks, such as terrain mapping, geological exploration, and marine observation. In recent years, new SAR sensors have acquired large amounts of SAR images. More importantly, the interval between targets being redetected is decreasing, and SAR data are updated more frequently than ever. Under this circumstance, SAR is more frequently applied to tasks with strict real-time requirements, such as disaster monitoring and battlefield reconnaissance.

SAR automatic target recognition (ATR), which aims to identify target categories in SAR images, is an essential research area in SAR image interpretation. Deep learning-based SAR ATR methods have shown obvious superiority over traditional methods due to its automatic data processing capability [1], [2]. These methods usually model the SAR ATR task as a classification problem in a closed set, where the training and the test samples share the same label space. Therefore, they are only applicable to ideal scenarios, where the training samples and the test samples are similarly distributed. However, real-world SAR ATR tasks need to be completed in open scenarios, where the acquisition of SAR data is an incremental process. Intuitively, the SAR ATR system has to process new classes at any point, especially when the system is applied to time-sensitive tasks, such as battlefield reconnaissance. Usually, it is difficult to collect enough data required for training an ideal classification model simultaneously. SAR may continuously capture echo signals from new classes of targets, which are unknown during the training stage and naturally misclassified into known categories. In order to ensure reliable recognition, traditional SAR ATR model has to be retrained with old classes and new classes together.

Although increasing training data is bound to improve the performance of SAR ATR models, repeated retraining of previous training samples will lead to significant time consumption and waste of storage resources, which is unconscionable for real-time SAR ATR tasks. Moreover, discarding the training samples of old targets directly and fine-tuning the trained model with new targets will cause catastrophic forgetting [3], in which the ability of recognizing old targets deteriorates. Therefore, it is necessary to develop new SAR ATR paradigm in open scenarios and design models that can quickly adapt to updated data and learn effective knowledge of new classes while maintaining the original classification capability. The SAR ATR model must strike a balance between processing efficiency and recognition performance; thus, it should be able to gradually adapt to the changes of sensor deployment environment and data distribution.

Class-incremental learning (CIL) is an advanced technique for target recognition in open scenarios. An intuitive explanation of CIL in open scenarios is illustrated in Fig. 1. The model

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Li Wang, Xinyao Yang, Haoyue Tan, and Feng Zhou are with the Ministry of Education, Key Laboratory of Electronic Information Countermeasure and Simulation, Xidian University, Xi'an 710071, China (e-mail: fzhou@mail.xidian.edu.cn).

Xueru Bai is with the National Laboratory of Radar Signal Processing, Xidian University, Xi'an 710071, China.

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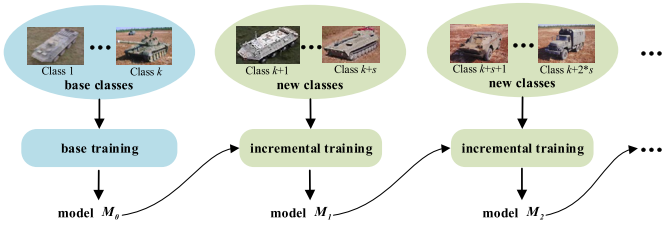


Fig. 1. Class-incremental learning starts with k base classes and the base model M_0. When a series of s new classes arrive sequentially, the model is updated to M_1, which is able to recognize $k+s$ classes. The model is updated continuously as new classes coming.

is first trained with base classes, and new classes can be added by providing corresponding training data. The model is then incrementally trained with new classes only, while possessing the capacity to recognize both base and new classes unitedly. Recent studies show that CIL-based SAR ATR methods can save computing resources and avert catastrophic forgetting on the premise of avoiding repeated training and storage of old classes. For instance, Dang et al. [4] proposed an open set SAR ATR model based on extreme value theory, Dang et al. [5] adjusted the decision boundary in open set recognition task according to the local geometrical and statistical information, and Tang et al. [6] proposed the high plastic error correction incremental learning method to alleviate model degradation and plasticity decline in the incremental SAR ATR task. The key of the existing CIL-based SAR ATR methods is to collect enough labeled training samples for new classes. However, in complex scenarios, the target distribution is extremely unbalanced and only a few samples can be collected for rare classes.

Few-shot learning (FSL) [7], as a small-instance deep learning paradigm, has been empirically shown effectiveness in limited data image recognition tasks. It aims to address the problems of inadequate generalization and inferior adaptability to rare classes of typical classification methods. It mitigates the severe performance degradation caused by the starvation of labeled samples. The available SAR ATR methods based on FSL can effectively tackle the issue of target recognition with few training samples in the closed set [8], [9], [10], [11]. However, it is difficult to apply them to SAR ATR tasks in the open scenario with new incoming targets.

Motivated by the abovementioned challenges, we are inspired to develop a new CIL-based SAR ATR method to adapt to new classes within a few shots in a computationally efficient and timely manner. The main challenge of SAR ATR in the few-shot class incremental learning (FSCIL) [12], [13] setting is that the starvation of labeled samples leads to severe overfitting to new targets and catastrophic forgetting of old targets.

In view of this, we propose a continually evolved SAR ATR model and design the corresponding training method. Specifically, we design a hierarchical embedding and incremental evolutionary network (HEIEN) based on a feature extractor and an evolved classifier embedded in a deep learning architecture. The main contributions of this article are summarized as follows.

1) A novel few-shot class-incremental SAR ATR framework with continuous evolutionary learning ability is proposed. By employing a graph attention network as the adaptive CIL (ACIL) module, we combine the knowledge learned on individual tasks and avoid forgetting old classes in CIL tasks.
2) A hybrid distance-based classifier is proposed to learn an embedding space with better separability. By constructing a metric that measures the similarity of SAR targets with Euclidean distance and cosine distance simultaneously, the samples possess better interclass difference and less interclass overlapping in the embedding space.
3) To enable effective model training with only a few labeled samples, a pseudo-incremental training method is proposed to train the learnable parameters in the framework. By repeated the sampling of pseudo-incremental tasks from the training set, the recognition capability can be generalized from old classes to new classes.
4) Experiments on the moving and stationary target acquisition and recognition (MSTAR) dataset show that the proposed method achieves satisfying recognition performance with remarkable advantages in few-shot class-incremental SAR target recognition tasks.

II. Related Work

A. Few-Shot Learning

FSL aims to extract the inductive bias from base classes and generalize it to new classes without the prohibitive cost of collecting large amounts of labeled samples. There are three mainstream paradigms on the current FSL methods: optimization-based methods, metric-based methods, and data augmentation methods. Optimization-based methods [14], [15] handle the problem in the learning to learn paradigm, where an optimizer controls the optimization of model parameters and try to enable fast model adaptation to few-shot data. Model-agnostic meta-learning (MAML) [16], [17] learns optimal initialization parameters for FSL tasks and optimize parameters continuously by generating simulation tasks. The metric-based methods map samples from the original space to the embedding space and then measure the similarity between samples according to some metric criteria, such as squared distance [18] and Earth mover's distance [19]. Data augmentation methods synthesize more data from the new classes to facilitate feature extraction. Zhang et al. [20] separate foregrounds and backgrounds of training samples and generate synthetic samples from various combinations of foreground and background slices. Hariharan and Girshick [21] generate extra training data by exploiting intraclass variation of the base classes. Deep generative models, such as generative adversarial networks [22], have also been applied to few-shot data augmentation. Yang et al. [23] generate more samples by estimating the distribution of new classes and calibrating the distributions of similar base classes.

B. Class-Incremental Learning

CIL aims at incrementally learning a unified classifier to recognize all classes encountered in a series of individual tasks. Whenever new data are added, the CIL model is not

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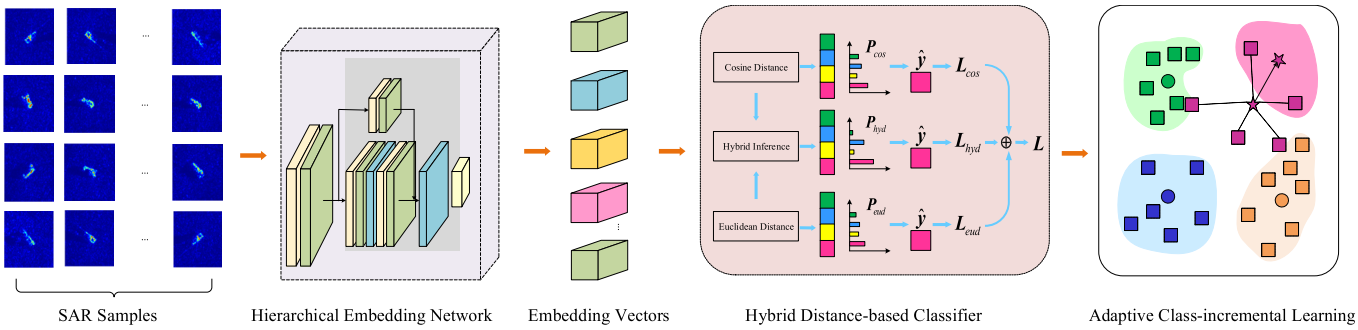


Fig. 2. Framework of HEIEN.

retrained but updated based on the samples of new classes and the original knowledge [24]. The main challenge of CIL is the catastrophic forgetting [25] on base classes when learning new ones. To solve the problem, much research has been carried out in the following two directions. The first direction is to preserve the important parameters learned from base classes, and the second one is to preserve the knowledge of the initial model. In the first direction, Li and Hoiem [26] use the data of new classes to fine-tune the initial model while maintaining the original capabilities. Kirkpatrick et al. [27] save the knowledge of base classes by selecting important weights. Zenke et al. [28] introduce biological complexity into CIL model by adding intelligent synapses that accumulates task-relevant information over time. In the second direction, distillation techniques are used to memorize important knowledge of base classes. Stacked gating autoencoders [29] are designed to learn a representation for the current task with only the new class data while maintaining the previous capabilities. Hou et al. [30] alleviate class-imbalance in CIL to reduce the ambiguities between old and new classes in the constraint of cosine normalization, less-forget constraint, and interclass separation. To avoid the high cost of collecting large amounts of data, some efforts have been devoted to CIL in few-shot scenarios. Tao et al. [31] propose a neural gas model to preserve the information of base classes and use label information to adjust feature space. Zhang et al. [32] use a graph model to control context information propagation between tasks. Zhao et al. [33] rebalance the information of base and new classes by regularization in the frequency domain. Dong et al. [34] construct a relation graph in base classes and mitigate forgetting via relation distillation.

Due to the distinct imaging mechanisms, SAR images usually have a lower resolution and are more susceptible to clutter than optical images. Limited by the starvation of training samples and unfixed target categories in FSCIL tasks, the separable features of different targets are more difficult to be extracted from SAR images than optical images. As a consequence, it is challenging for the existing FSCIL methods to achieve satisfying performance on SAR ATR tasks directly.

III. Proposed Method

This section introduces the proposed few-shot class-incremental SAR ATR method named HEIEN that consists of three main stages, as shown in Fig. 2. In the feature embedding stage, a hierarchical embedding network consisting of stacked residual modules is utilized to map SAR samples to embedding vectors. The hierarchical embedding network is trained with only base classes. Then, a hybrid distance-based classifier is designed to predict unlabeled samples. Finally, an ACIL strategy is proposed to propagate context information between base classes and new classes and constrain the classifiers to continually evolve on a series of new tasks. The detailed components and training process of HEIEN are described as follows.

A. Problem Definition

For a typical FSCIL task, the model learning process can be divided into several recognition tasks that come in sequence. Let $\{\mathcal{T}^0, \mathcal{T}^1, \ldots, \mathcal{T}^n\}$ denotes the stream of incremental recognition tasks, and $\{\mathbb{Z}^0, \mathbb{Z}^1, \ldots, \mathbb{Z}^n\}$ and $\{\mathbb{Q}^0, \mathbb{Q}^1, \ldots, \mathbb{Q}^n\}$ are the corresponding labeled training sets and unlabeled test sets. $\mathbb{Z}^n = \{(x_i, y_i)\}_{i=1}^{I_n}$, where $y_i \in \mathbb{C}^n$, I^n is the number of training samples in $\mathbb{Z}^n$, and $\mathbb{C}^n$ is the label space of $\mathbb{Z}^n$. It should be noted that the label space of different tasks is not overlapped, i.e., $\forall j, k, \mathbb{C}^j \cap \mathbb{C}^k = \emptyset$. Except that the first task $\mathcal{T}^0$ has sufficient labeled training samples, the following tasks have only a few training samples in each class. For a specific incremental task $\mathcal{T}^n$, $\mathbb{Z}^n$ can be described as a C-way K-shot dataset, where there are C classes of targets in $\mathbb{Z}^n$, and there are K labeled training samples in each class. In the current task, the training samples in previous tasks cannot be reused for training, while the test samples in each task belong to classes in all previous tasks and the current task. The classes contained in $\mathcal{T}^0$ are also called base classes. The overall goal of the above learning process is to train a model with training samples in the base classes, and the model can continue to learn knowledge from the new classes in the subsequent incremental tasks although there are only a few labeled training samples of each new class.

B. Hierarchical Embedding Network

As shown in the framework, each sample is mapped from the original SAR space to a new embedding space by a hierarchical embedding network, which is equivalent to an embedding function $f_\varphi : \mathbb{R}^D \to \mathbb{R}^L$ with learnable parameters φ. In this process, each sample is extracted typical features and mapped into an L-dimensional vector by the embedding network with a stacked architecture. Fig. 3 shows the

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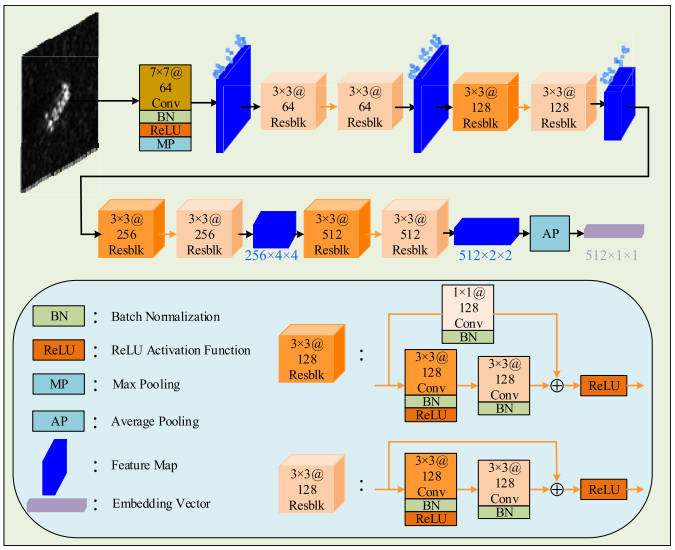


Fig. 3. Structure of the hierarchical embedding network.

structure of the embedding network that consists of several convolutional blocks and an average pooling layer. The first convolutional block is the traditional block, which consists of the common batch normalization layer, the rectified linear unit (ReLU) activation function, and the max pooling layer. The rest convolutional blocks are two kinds of residual blocks.

C. Hybrid Distance-Based Classifier

The core to construct an effective classifier with only a few training samples is designing an appropriate metric. The existing Euclidean distance-based classifier is not suitable for FSCIL SAR ATR tasks due to its defects in robustness and generalization. Once the feature vector of each SAR sample is learned by the hierarchical embedding network, the decision boundary needs to be divided by the classifier in the feature space. Motivated by the obvious fact that a good classifier should constrain SAR targets of different classes to have smaller intraclass distance and larger interclass distance in feature space and makes full use of the supervision information of few labeled samples, we design a hybrid distanced-based classifier that combines the superiority of two different metrics.

For a C-class classification problem in an arbitrary recognition task $\mathcal{T}^n$, the labeled training set $\mathbb{Z}^n = \{(x_i, y_i)\}_{i=1}^{N_n}$ is given and $y_i \in \{1, \ldots, C\}$ is the corresponding label. The clustering center $w_c^n \in R^L (c \in 1, 2, \ldots, C)$ of each class in task $\mathcal{T}^n$ can be calculated by

$$w_c^n = \frac{1}{\left|\mathbb{Z}_c^n\right|} \sum_{(x_i, y_i) \in \mathbb{Z}_c^n} f_\phi(x_i) \tag{1}$$

where $\mathbb{Z}_c^n$ denotes the set of labeled samples with class c in $\mathbb{Z}^n$.

First, two subclassifiers with similar mathematical form to the Softmax classifier are constructed to measure the similarity of SAR targets in Euclidian and cosine space, respectively,

$$p_{\text{Euc}}^n(y_i = c|x_i) = \frac{\exp\left(-d_{\text{Euc}}\left(f_\phi(x_i), w_c^n\right)\right)}{\sum_c^C \exp\left(-d_{\text{Euc}}\left(f_\phi(x_i), w_c^n\right)\right)} \tag{2}$$

$$p_{\cos}^n(y_i = c|x_i) = \frac{\exp\left(-d_{\cos}\left(f_\phi(x_i), w_c^n\right)\right)}{\sum_c^C \exp\left(-d_{\cos}\left(f_\phi(x_i), w_c^n\right)\right)} \tag{3}$$

where $p_{\text{Euc}}^n(y_i = c|x_i)$ and $p_{\cos}^n(y_i = c|x_i)$ are the confidence of x_i being classified to the kth category in task $\mathcal{T}^n$; $d_{\text{Euc}}(\cdot)$ and $d_{\cos}(\cdot)$ are the Euclidean distance and cosine distance, respectively.

Then, the hybrid distance-based classifier is obtained by combining the classification results of the two subclassifiers according to their confidence

$$\begin{aligned} p_{\text{Hyb}}^n(y_i = c|x_i) &= \alpha^n \cdot p_{\text{Euc}}^n(y_i = c|x_i) + \beta^n \cdot p_{\cos}^n(y_i = c|x_i) \\ &= \frac{\alpha^n \cdot \exp\left(-d_{\text{Euc}}\left(f_\varphi(x_i), w_c^n\right)\right)}{\sum_c^C \exp\left(-d_{\text{Euc}}\left(f_\varphi(x_i), w_c^n\right)\right)} \\ &\quad + \frac{\beta^n \cdot \exp\left(-d_{\cos}\left(f_\varphi(x_i), w_c^n\right)\right)}{\sum_c^C \exp\left(-d_{\cos}\left(f_\varphi(x_i), w_c^n\right)\right)} \end{aligned} \tag{4}$$

where α^n and β^n are the parameters to automatically adjust the mixing ratio the two subclassifiers

$$\alpha^n = \frac{\arg\max_{c \in [0,C]} \alpha_c^n}{\arg\max_{c \in [0,C]} \alpha_c^n + \arg\max_{c \in [0,C]} \beta_c^n} \tag{5}$$

$$\beta^n = \frac{\arg\max_{c \in [0,C]} \beta_c^n}{\arg\max_{c \in [0,C]} \alpha_c^n + \arg\max_{c \in [0,C]} \beta_c^n} \tag{6}$$

where $\alpha_c^n (c \in 1, 2, \ldots, C)$ and $\beta_c^n (c \in 1, 2, \ldots, C)$ are the soft labels of class C given by p_{Euc}^n and $p_{\cos}^n$.

As a matter of course, the classifiers mathematically similar to softmax classifier can be trained by the cross-entropy loss function. In order to constrain the hierarchical embedding network to learn an embedding space where both Euclidean distance and cosine distance perform well as the effective similarity metric, an enhanced hybrid loss is proposed as the objective function. For test samples $\{(x_i, y_i), i = 1, \ldots, m\}$ in each training episode of $\mathcal{T}^n$, the objective loss can be calculated by

$$\begin{aligned} L_{\text{Enhanced}}^n = &-\frac{1}{m} \sum_i \log p_{\text{Euc}}^n(y = y_i|x_i) \\ &-\frac{1}{m} \sum_i \log p_{\cos}^n(y = y_i|x_i) \\ &-\frac{1}{m} \sum_i \log p_{\text{Hyb}}^n(y = y_i|x_i). \end{aligned} \tag{7}$$

By adding the cross-entropy loss of p_{Euc}^n and $p_{\cos}^n$ directly to the final loss, the interclass differences of samples can be directly constrained by Euclidean distance and cosine distance in the embedding space.

D. ACIL Module

As SAR samples come with groups in the FSCIL task, the classifier learned on individual incremental recognition tasks can only provide credible recognition results for the current task. The information learned in previous tasks is gradually

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forgotten as the learning tasks increase and it is difficult to maintain reliable discrimination ability for all previous classes of targets. Therefore, it is necessary to design a continual learning mechanism to prevent forgetting the context information learned from all previous tasks. As introduced above, the core of the hybrid distanced-based classifier is to calculate the appropriate clustering center of each class, especially the new class with few labeled samples in the incremental classification task. Motivated by these considerations, an ACIL module is proposed to update the clustering center according to the global context information.

Let $\mathbf{W}^n \in \mathbb{R}^{C_n \times L}$ denotes the clustering center matrix in the hybrid distance-based classifier learned on task $\mathcal{T}^n$, where each row $\vec{w}_c^n$ in $\mathbf{W}^n$ is the clustering center of a specific class c, C_n is the total number of classes in the task $\mathcal{T}^n$, and L is the dimension of embedding vector. To refine the discriminability of the classifier in the current task $\mathcal{T}^n$, we can adjust $\vec{w}_c^n$ by propagate information from $\vec{w}_i^n$ of all other classes. First, the clustering center of all classes in all the tasks up to now can be collected as $\mathbf{W}^n = [\vec{w}_1^0, \vec{w}_2^0, \ldots, \vec{w}_1^i, \vec{w}_2^i, \ldots, \vec{w}_{C_n}^n]$.

A graph attention network [35] g_θ with learnable parameters θ is utilized to model the relationship between all clustering centers in W^N. Each clustering center can be regarded as a node in the graph and receive context information from other nodes. The attention mechanism is utilized to propagate information between nodes. At a SAR ATR task, the fully connected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consists of nodes $\mathcal{V}? = \{\vec{w}_j\}_{j=1}^M$ and edges $\mathcal{E}? = \{e_{jk}\}_{j,k=1}^M (j \neq k)$, where $\vec{w}_j$ is the jth node, e_{jk} is the edge connecting $\vec{w}_j$ and $\vec{w}_k$, and M is the total number of nodes.

For any two nodes $\vec{w}_j$ and $\vec{w}_k$, the relation coefficient e_{jk} can be calculated by

$$e_{jk} = \lambda \cdot \sigma\left(d_{\cos}\left(\psi\left(\vec{w}_j\right), \psi\left(\vec{w}_k\right)\right)\right) + (1-\lambda) \cdot \sigma\left(d_{\text{Euc}}\left(\psi\left(\vec{w}_j\right), \psi\left(\vec{w}_k\right)\right)\right) \quad (8)$$

where ψ is a liner transformation functions that project clustering centers to a new space, $\sigma(\cdot)$ is the sigmoid function, and λ is the coefficient that adjusts the ratio of cosine similarity and Euclidean similarity. The task indexes in the superscript of $\vec{w}_j$ and $\vec{w}_k$ are omitted for clarity. The attention weights of the node j can be calculated by normalizing the all the coefficients with the softmax function

$$a_{jk} = \text{soft}\max\left(e_{jk}\right) = \frac{\exp\left(e_{jk}\right)}{\sum_{h=1}^{|\mathbf{W}|} \exp\left(e_{jh}\right)}. \quad (9)$$

Then, each node in the graph can be updated by aggregating information from all the nodes based on the attention weights

$$\vec{w}_j' = \vec{w}_j + \left(\sum_{k=1}^{|\mathbf{W_n}|} a_{jk} \mathbf{U} \vec{w}_k\right) \quad (10)$$

where $\mathbf{U}$ is a learnable linear transformation matrix corresponding to the value matrix in the self-attention mechanism [36]. All nodes in the graph are updated by the above operations and the modified clustering centers can be obtained.

In each incoming task, the ACIL module is utilized to update the clustering centers learned in the current task and previous task. Finally, the clustering centers of current task can be obtained by concatenating the incrementally learned clustering centers over tasks from the past to the present.

E. Pseudo-Incremental Training

In the proposed framework, both the hierarchical embedding network f_φ and the ACIL module g_θ have their own learnable parameters. To avoid catastrophic forgetting, a decoupled training method is designed to learn them, respectively. First, f_φ is trained on the base task $\mathcal{T}^0$ in the same way as training a traditional convolutional neural network (CNN) and the learned parameters are fixed in the incremental task $\mathcal{T}^n (n \geq 1)$.

For effective context information encoding in the ACIL module, it is important to learn g_θ in the few-shot class-incremental settings. However, the starvation of labeled data hinders learning effective θ. To solve this problem, a pseudo-incremental training method is proposed with the inspiration from meta-learning. Pseudo-incremental tasks are constructed by randomly sampling from the base training set $\mathbb{Z}^0$ to mimic the few-shot incremental testing scenario. In each pseudo-incremental training episode, some classes are randomly selected from $\mathbb{Z}^0$ as the pseudo base classes, while the rest are selected as the pseudo-incremental classes. For both the pseudo-incremental classes and pseudo-base classes in the ith epoch, the corresponding training set and test set are denoted by $\{\mathbb{Z}_B^i, \mathbb{Q}_B^i\}$ and $\{\mathbb{Z}_I^i, \mathbb{Q}_I^i\}$, respectively. The training sets are used to compute the clustering centers and the test sets are used to compute objective loss for optimizing θ. At the end of the iteration, θ is frozen and the ACIL module is learned. The procedure of pseudo-incremental training is illustrated as Algorithm 1.

Algorithm 1 Pseudo Incremental Training Under Few-Shot Class-Incremental Scenarios

Input: The base training set $\mathbb{Z}^0$; randomly initialized ACIL module g_θ; hierarchical embedding network f_φ; training epochs R.
Output: Trained ACIL module g_θ.

```
i ← 0; R ← trainingepochs; ▷ Initialization
while i < R do
    Generate the training and test set {Z_B^i, Q_B^i} for pseudo base classes from Z^0 by random sampling;
    Generate the training and test set {Z_I^i, Q_I^i} for pseudo incremental classes from Z^0 by random sampling;
    W^n ← Learn clustering centers upon f_φ with Z_B^i and Z_I^i;
    W^n' ← Update W^n with the ACIL module g_θ;
    L_Enhanced^i ← Compute objective loss with f_φ, W^n', Q_B^i and Q_I^i;
    θ' ← Update θ with stochastic gradient descent
    i = i + 1; ▷ Update the current epoch
end while
```

IV. Experiment

In this section, experimental results will be given to the performance of the proposed method. The experiments require

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TABLE I
TRAINING SET AND TEST SET FOR THE EXPERIMENTAL SETUP

		Training set		Test set	
Task	Class	Depression	No.	Depression	No.
0	BTR60	17°	256	15°	195
	T72	17°	691	15°	582
	2S1	17°	299	15°	274
	T62	17°	299	15°	273
	ZIL131	17°	299	15°	274
	ZSU234	17°	299	15°	274
1	BRDM2	17°	K=1, 3, 5	15°	274
2	D7	17°	K=1, 3, 5	15°	274
3	BMP2	17°	K=1, 3, 5	15°	587
4	BTR70	17°	K=1, 3, 5	15°	196

powerful computing resources, and they are conducted on the platform with the NVIDIA RTX 3090 GPU. The implementation of neural networks relies on the machine learning framework PyTorch 2.0.

A. Dataset

The performance of the proposed method will be verified by experiments in the MSTAR [37] dataset, which includes ten classes of military vehicles, i.e., BTR60, T72, 2S1, T62, ZIL131, ZSU234, BRDM2, D7, BMP2, and BTR70. The MSTAR dataset consists of targets at the depression of 15° and 17° under standard operating conditions. As a matter of routine, the training set and the test set in this article include the targets at the depression angle of 17° and 15°, respectively.

For the FSCIL problem in this article, the dataset needs to be divided into several labeled datasets $\{\mathbb{Z}^0, \mathbb{Z}^1, \ldots, \mathbb{Z}^n\}$ and unlabeled datasets $\{\mathbb{Q}^0, \mathbb{Q}^1, \ldots, \mathbb{Q}^n\}$ in the series of tasks $\{\mathcal{T}^0, \mathcal{T}^1, \ldots, \mathcal{T}^n\}$. Six classes of targets are included in $\mathcal{T}^0$ as the base classes, i.e., BTR60, T72, 2S1, T62, ZIL131, and ZSU234. The rest four classes of targets serve as the incremental classes in the $\mathcal{T}^n (n \geq 1)$. Specifically, K ($K = 1$, 3, and 5 in this article) samples of each incremental class at 17° are randomly selected as $\mathbb{Z}^n (n \geq 1)$ in each incremental task. In this article, each incremental task includes an extra incremental class compared to the previous task. Detailed information of the training set and the test set is listed in Table I. It should be noted that the model parameters are only learned in the base task $\mathcal{T}^0$. Although the parameters are fixed in the incremental recognition task, different training samples lead to different clustering centers, which will influence the classification performance.

B. Recognition Results of Base Task

In the proposed framework, the model first needs to be trained on in the base task $\mathcal{T}^0$, that is the same as the traditional SAR ATR task. To further illustrate the effectiveness of the hybrid distance-based classifier, three classifiers based on cosine distance, Euclidian distance, and hybrid distance were

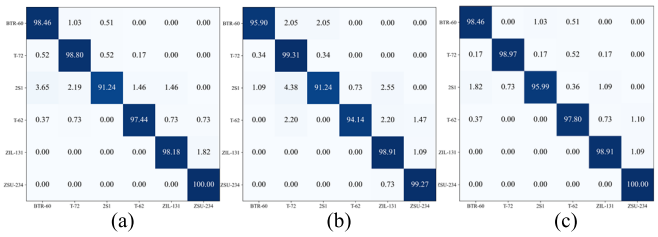


Fig. 4. Confusion matrixes of different classifiers in the base task $\mathcal{T}^0$. (a) Cosine classifier. (b) Euclidian classifier. (c) Hybrid distance-based classifier.

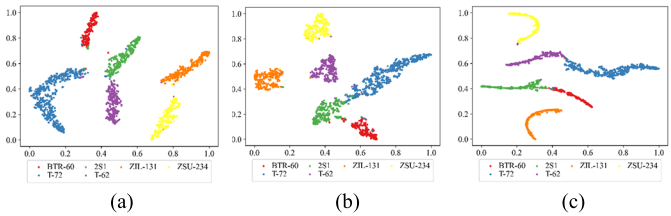


Fig. 5. Visualization of the embedding vectors of different classifiers in the base task $\mathcal{T}^0$. (a) Cosine classifier. (b) Euclidian classifier. (c) Hybrid distance-based classifier.

TABLE II
RECOGNITION PERFORMANCE OF DIFFERENT CLASSIFIERS IN THE BASE TASK $\mathcal{T}^0$

Classifier	Cosine	Euclidian	Hybrid
Accuracy/%	97.54	96.96	98.45

employed as comparative experiments. All classifiers use the same hierarchical embedding network for feature extraction. The recognition performance of different classifiers in the base task $\mathcal{T}^0$ is shown in Table II. It is observed that the hybrid distance-based classifier outperforms the other two classifiers based on single distance. Because there are enough training samples, the recognition accuracy of all classifiers in the basic task $\mathcal{T}^0$ exceeds 95%. The corresponding confusion matrixes of the three classifiers are shown in Fig. 4. In particular, the recognition accuracy of 2S1 output by the cosine classifier or the Euclidian classifier is only 91.24%, which is obviously lower than that of the hybrid distance-based classifier. The above results show that the hybrid distance-based metric can effectively constrain the hierarchical embedding network to learn embedding vectors with better separability. In order to explain the effectiveness of the hybrid distance-based classifier more intuitively, the visualization of embedding vectors by t-distributed stochastic neighbor embedding (t-SNE) [38] is shown in Fig. 5. Compared with the other two classifiers, the distribution of embedding vectors belongs to the same target is more concentrated, which makes the reasonable decision boundaries can be divided more easily.

C. Recognition Results of FSCIL Task

In this section, the effectiveness of the proposed method is illustrated by the ablation studies of each module in HEIEN, and the comparison results are shown in Table III. In order to study the effect of the number of labeled samples on the

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TABLE III
RECOGNITION ACCURACY OF DIFFERENT CLASSIFIERS IN FEW-SHOT CLASS-INCREMENTAL SAR ATR TASKS

Classifier	Shot	Accuracy in each class-incremental task/%			
		T^1	T^2	T^3	T^4
Cosine	1shot	85.55	75.87	62.55	58.41
	3shot	89.84	81.16	66.98	62.63
	5shot	89.98	81.90	70.93	67.91
Euclidian	1shot	83.88	74.67	63.78	59.66
	3shot	88.63	80.54	68.87	65.22
	5shot	90.17	82.44	70.20	66.38
Hybrid	1shot	88.16	80.83	68.47	63.47
	3shot	89.93	82.23	70.40	66.78
	5shot	91.75	83.80	72.80	69.00
HEIEN (Hybrid + ACIL)	1shot	91.15	86.94	76.55	73.03
	3shot	91.75	88.10	80.08	75.68
	5shot	93.85	90.54	83.27	82.05

TABLE IV
PERFORMANCE IMPROVED BY ACIL MODULE IN FEW-SHOT CLASS-INCREMENTAL SAR ATR TASKS

Shot	T^1	T^2	T^3	T^4
1shot	2.99	6.11	8.08	9.56
3shot	1.82	5.87	9.68	8.90
5shot	2.10	6.74	10.47	13.05

few-shot class-incremental SAR ATR task, the number of shots in each task is set to 1, 3, and 5, respectively. It can be seen that the recognition performance improves as the number of shots increases in each classifier. When the training samples are scarce, even adding a few training samples is of great benefit to the improvement of recognition performance. Similar to the basic task $\mathcal{T}^0$, the recognition accuracy of the hybrid distance-based classifier is also obviously higher than that of the rest single distance-based classifiers. To further illustrate the effectiveness of the ACIL module, the recognition performance improved by adding the ACIL module is shown in Table IV. It can be seen that when there are more new targets in the incremental task, the ACIL module plays a more obvious role in overcoming forgetting the knowledge of old targets. In the 1shot, 3shot, and 5shot settings of the incremental task $\mathcal{T}^4$, the ACIL module can improve the accuracy by 9.56%, 8.90%, and 13.05%, respectively, which indicates its strong applicability in open scenarios. In general, with the increase of incremental tasks, more few-shot targets need to be classified together with all targets in previous tasks, and the recognition accuracy will decrease significantly. To a certain extent, the ACIL module can prevent the decline of recognition accuracy, and the effect is more obvious in the later tasks.

In order to illustrate the effect of hybrid distance-based classifier and the ACIL module in more detail, the confusion matrixes of different classifiers in 5shot settings are shown in Fig. 6. Fig. 6(a)–(p) represents the confusion matrixes of different classifiers in $\mathcal{T}^1$–$\mathcal{T}^4$. For each confusion matrix, the first six targets are from the base task, and the rest are from incremental recognition tasks. In tasks $\mathcal{T}^1$–$\mathcal{T}^4$, there are one to four classes of incremental targets.

TABLE V
PERFORMANCE OF DIFFERENT CLASSIFIERS ON THE BASE AND INCREMENTAL CLASSES

Classifier	Classes	Accuracy/%			
		T^1	T^2	T^3	T^4
Cosine	Base	97.17	88.57	87.88	85.63
	Incremental	40.88	59.13	43.00	42.98
Euclidean	Base	92.90	84.83	78.69	77.99
	Incremental	71.53	74.27	56.21	50.04
Hybrid	Base	96.96	95.14	92.84	88.52
	Incremental	56.20	45.08	39.73	41.55
HEIEN	Base	94.66	93.22	89.91	89.26
	Incremental	88.32	81.39	72.34	71.90

In order to analyze the recognition performance of different classifiers more intuitively, the recognition performance of different classifiers on the base and incremental classes in the 5shot setting is shown in Table V, respectively. It can be seen that the hybrid distance-based classifier has the highest recognition accuracy on the base classes but does not perform well on the incremental classes. In most incremental tasks, the recognition performance of hybrid distance-based classifier is even lower than that of the cosine classifier or the Euclidean classifier on incremental classes. It indicates that the hybrid distance-based classifier suffered from severe overfitting on the base classes. In other words, the classifier is more inclined to rely on the old knowledge learned from base classes and cannot adapt to the new knowledge in incremental tasks effectively. Compared with the hybrid distance-based classifier, the recognition accuracy of HEIEN on incremental classes in $\mathcal{T}^1$–$\mathcal{T}^4$ improved by about 30% after adding the ACIL module. Meanwhile, the recognition accuracy of HEIEN on the base classes decreases by about 2%–3% compared with that of the hybrid distance-based classifier on $\mathcal{T}^1$–$\mathcal{T}^3$ and increases by 0.74% on $\mathcal{T}^4$. This demonstrates that HEIEN can learn useful information of incremental classes while preserving more knowledge of base classes by integrating the knowledge from previous classifiers.

D. Comparison Between HEIEN and Other Incremental Learning Methods

To adequately demonstrate the superiority of the proposed HEIEN, we will compare the performance of HEIEN with the existing advanced incremental learning methods, including the angular loss incremental classification (ALICE) [39], constrained FSCIL (C-FSCIL) [40], forward compatible training (FACT) [41], and self-supervised stochastic classifier (S3C) [42], as shown in Table VI. It can be seen that HEIEN achieves the highest recognition accuracy regardless of the 1shot, 3shot, or 5shot class-incremental SAR ATR scenarios.

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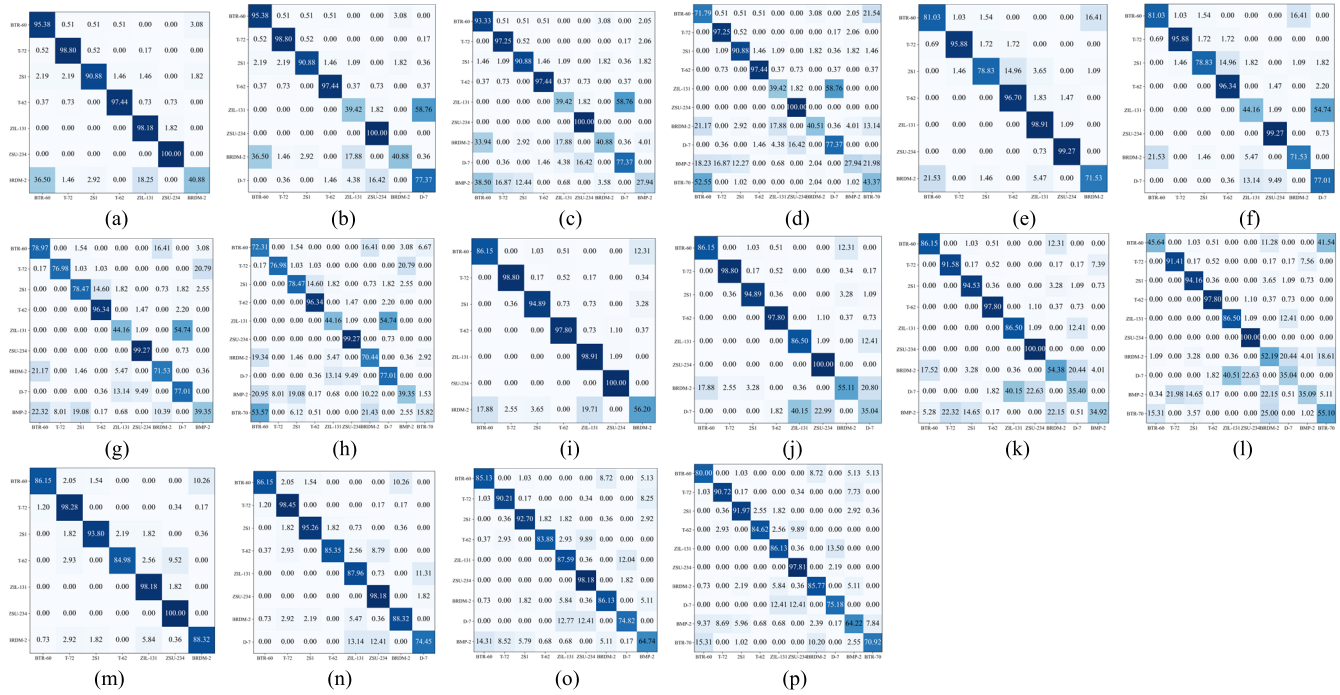


Fig. 6. Confusion matrixes of different classifiers in 5shot settings. (a)–(d) Cosine classifier in $\mathcal{T}^1$–$\mathcal{T}^4$. (e)–(h) Euclidian classifier in $\mathcal{T}^1$–$\mathcal{T}^4$. (i)–(l) Hybrid distance-based classifier in $\mathcal{T}^1_\mathcal{T}^4$. (m)–(p) HEIEN in $\mathcal{T}^1$–$\mathcal{T}^4$.

TABLE VI

COMPARISON BETWEEN HEIEN AND OTHER INCREMENTAL LEARNING METHODS

Shot	Methods	Accuracy in each class-incremental task/%			
		T^1	T^2	T^3	T^4
1shot	ALICE	81.20	75.17	58.76	57.29
	C-FSCIL	88.68	80.21	64.45	61.35
	FACT	90.26	82.95	72.98	69.21
	S3C	84.20	82.11	69.44	61.54
	HEIEN	**91.15**	**86.94**	**76.55**	**73.03**
3shot	ALICE	82.25	77.44	64.45	61.22
	C-FSCIL	91.52	84.71	70.30	67.59
	FACT	90.90	83.32	73.82	70.03
	S3C	89.33	83.02	78.92	73.40
	HEIEN	**91.75**	**88.10**	**80.08**	**75.68**
5shot	ALICE	83.78	77.98	65.78	62.72
	C-FSCIL	92.50	85.66	76.55	73.90
	FACT	93.06	84.24	77.31	75.44
	S3C	90.54	86.78	80.31	77.33
	HEIEN	**93.85**	**90.54**	**83.27**	**82.05**

Due to its excellent continuous evolutionary learning ability, the performance of HEIEN on few shot class-incremental SAR ATR tasks is significantly superior to other methods.

E. Superiority of the Adaptive Mixing Ratio in Hybrid Distance-Based Classifier

The adaptive mixing strategy is used in HEIEN to automatically calculate the mixing ratio according to the confidence of two individual classifiers. To further establish the superiority of the proposed strategy, the accuracy of the classifiers based on adaptive mixing ratio and different fixed ratios on incremental tasks in the 5shot setting is shown in Table VII. In total, there are nine different fixed ratio settings and the mixing ratio is between 0.1 and 0.9. No matter how the mixing ratio is adjusted, the recognition accuracy of the classifiers based on fixed ratio is lower than that based on adaptive ratio. This proves that the proposed adaptive mixing strategy can combine the advantages of cosine classifier and Euclidean classifier and calculate the optimal mixing ratio on each sample.

TABLE VII

ACCURACY OF HYBRID DISTANCE-BASED CLASSIFIER WITH DIFFERENT MIXING RATIOS IN 5SHOT CLASS-INCREMENTAL TASKS

Mixing Ratio		Accuracy/%			
Cosine	Euclidian	T^1	T^2	T^3	T^4
0.1	0.9	90.31	82.81	70.27	66.84
0.2	0.8	90.35	82.44	70.34	66.59
0.3	0.7	90.17	82.31	70.47	66.75
0.4	0.6	90.03	82.73	70.54	66.91
0.5	0.5	90.45	82.36	70.90	67.25
0.6	0.4	90.49	82.64	70.64	67.34
0.7	0.3	90.03	82.40	71.13	67.31
0.8	0.2	89.89	82.19	70.97	67.66
0.9	0.1	89.84	81.98	70.83	67.87
Adaptive		91.75	83.80	72.80	69.00

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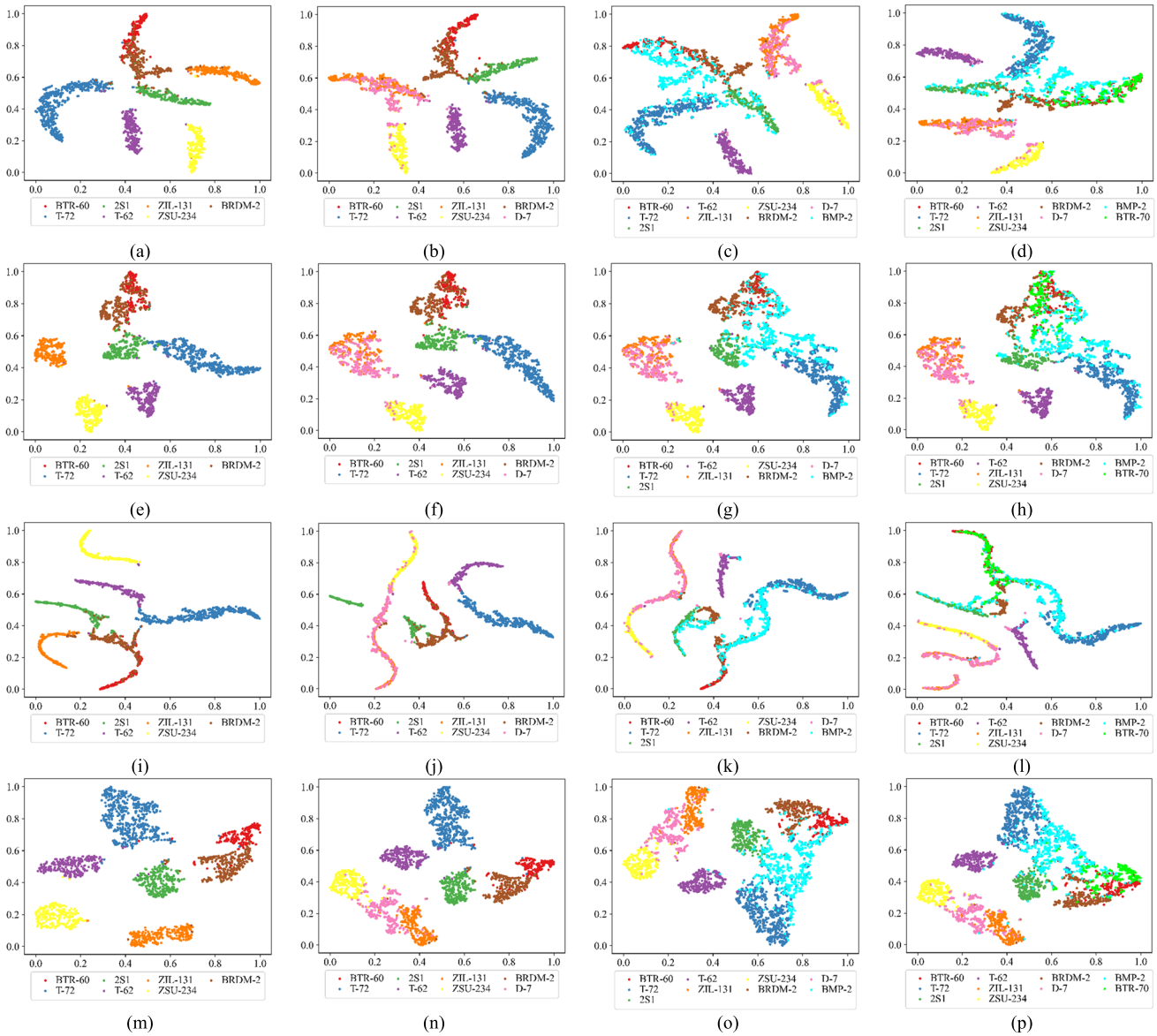


Fig. 7. Visualization of the embedding vectors of different classifiers in 5shot settings. (a)–(d) Cosine classifier in $\mathcal{T}^1$–$\mathcal{T}^4$. (e)–(h) Euclidian classifier in $\mathcal{T}^1$–$\mathcal{T}^4$. (i)–(l) Hybrid distance-based classifier in $\mathcal{T}^1$–$\mathcal{T}^4$. (m)–(p) HEIEN in $\mathcal{T}^1$–$\mathcal{T}^4$.

F. Visualization and Interpretability of HEIEN

In this section, the effectiveness and interpretability of the proposed method in few-shot incremental SAR ATR will be further illustrated by t-SNE. All of the visualization results are achieved in 5shot settings. Because the cosine classifier only considers the angle between embedding vectors, the distribution of samples belong to the same class will be elongated in the embedding space. On the contrary, the Euclidean classifier mainly considers the absolute distance between embedding vectors, so the corresponding visualization results are closer to blocky. The hybrid distance-based classifier tends to learn an embedding space suffered from severe overfitting. The distribution of embedding vectors in this space is narrower than that of both cosine classifier and Euclidean classifier, which leads to more severe interclass aliasing. It can be seen from Fig. 7(l) that the aliasing degree of BRDM-2 and BTR-70 in the embedding space of the hybrid distance-based classifier is obviously higher than that of other classifiers. As shown in the third and fourth row of Fig. 7, the interclass separability of embedding vectors in the embedding space of HEIEN is significantly better than that of the hybrid distance-based classifier, which illustrates that the ACIL module can alleviate overfitting by propagating information learned from different tasks. Moreover, the embedding space learned by HEIEN can achieve an ingenious balance between the intraclass unity and interclass difference, so as to improve the separability of incremental classes in the embedding space while maintaining that of base classes.

V. Conclusion

Existing SAR ATR methods have insurmountable flaws in identifying new targets with few training samples in the open scenario. In this article, the HEIEN is proposed to solve the problem of few-shot FSCIL in SAR ATR. By taking a

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hybrid distance-based classifier, HEIEN makes effective use of the supervised information in the cosine and Euclidean space simultaneously. Then, an ACIL module is designed to combine the knowledge learned on individual SAR ATR tasks and avoid catastrophic forgetting of old classes. In addition, a pseudo-incremental training strategy is proposed to ensure the effectiveness of model training. However, with the increase of incremental classes, the recognition accuracy of the classifier will gradually decrease. The authors will try to solve this problem in the future work.

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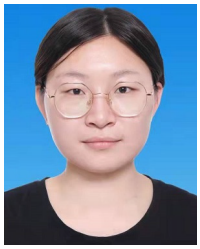
Li Wang was born in Jiangsu, China, in 1992. He received the B.S. and Ph.D. degrees from Xidian University, Xi'an, China, in 2015 and 2020, respectively.

He is currently a Lecturer with the Ministry of Education, Key Laboratory of Electronic Information Countermeasure and Simulation, Xidian University. His major research interests include deep learning and radar automatic target recognition (ATR).



Xinyao Yang was born in Jiangxi, China, in 1999. He received the B.S. degree from Nanchang University, Jiangxi, in 2021. He is currently pursuing the M.S. degree with Xidian University, Xi'an, China.

His major research interests include deep learning and radar automatic target recognition (ATR).



Haoyue Tan was born in Hebei, China, in 1999. She received the B.S. degree from Harbin Engineering University, Harbin, China, in 2021. She is currently pursuing the M.S. degree with Xidian University, Xi'an, China.

Her major research interests include deep learning and cognitive radio.



Xueru Bai (Senior Member, IEEE) was born in Xi'an, Shaanxi, China, in 1984. She received the B.S. and Ph.D. degrees in signal and information processing from Xidian University, Xi'an, in 2006 and 2011, respectively.

She is currently a Professor with the National Laboratory of Radar Signal Processing, Xidian University. Her research interests include high-resolution radar imaging and radar automatic target recognition (ATR).

Dr. Bai was a recipient of the National Excellent Doctoral Dissertation Award granted by the Ministry of Education of China, and the Program for Excellent Young Scientist selected by the National Natural Science Foundation of China.



Feng Zhou (Member, IEEE) was born in Tongxu, Henan, China, in 1980. He received the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 2007.

He is currently a Director and a Professor with the Key Laboratory of Electronic Information Countermeasure and Simulation Technology, Ministry of Education, Xidian University. He has authored or coauthored over 80 articles. His research interests include high-resolution radar imaging and radar countermeasures.

Dr. Zhou was a recipient of the Young Scientist Award from the XXXI URSI GASS Committee, the program for Support of Top-Notch Young Professionals from the Central Organization Department of China, and the Program for New Century Excellent Talents in university from the Ministry of Education of China.

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